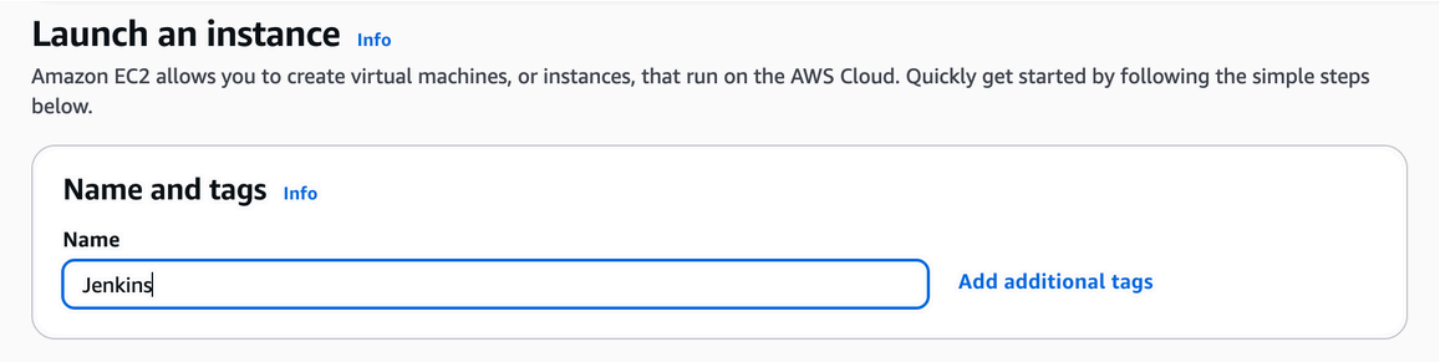
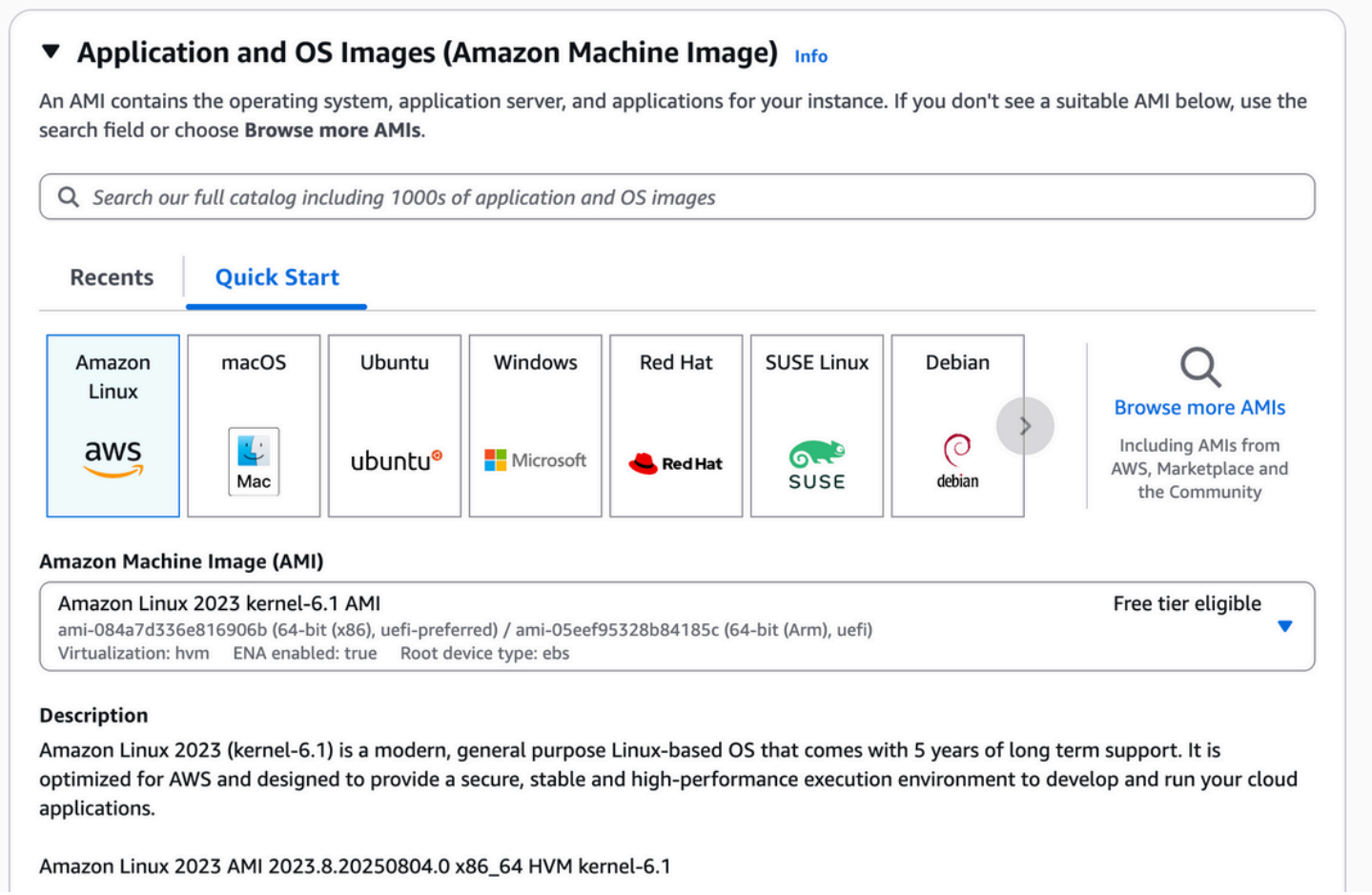
JENKINS DEPLOYMENT

STEP-1: LAUNCH 1 INSTANCE FOR JENKINS

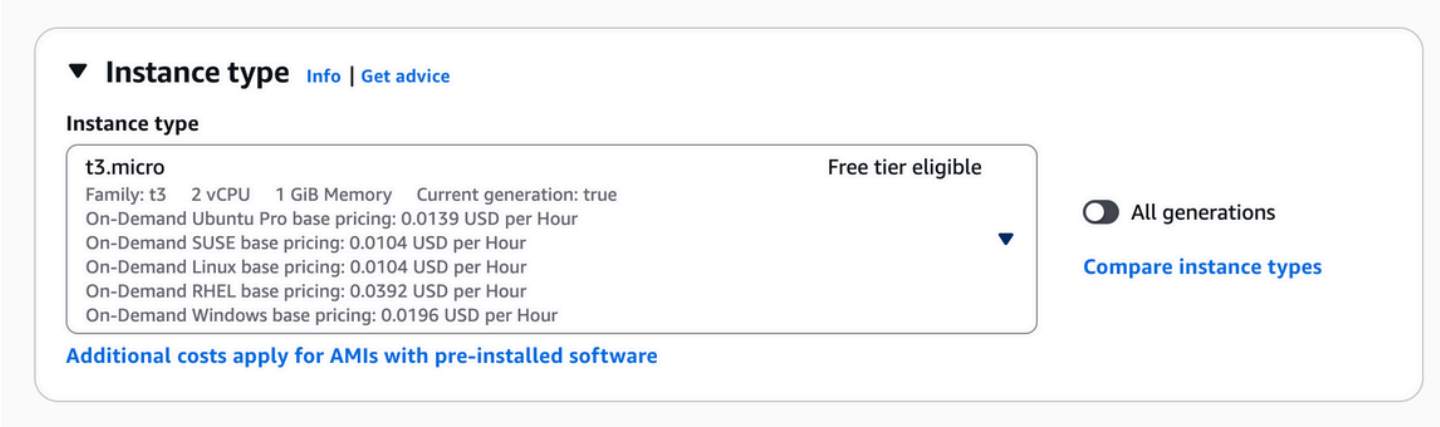
Name : Jenkins



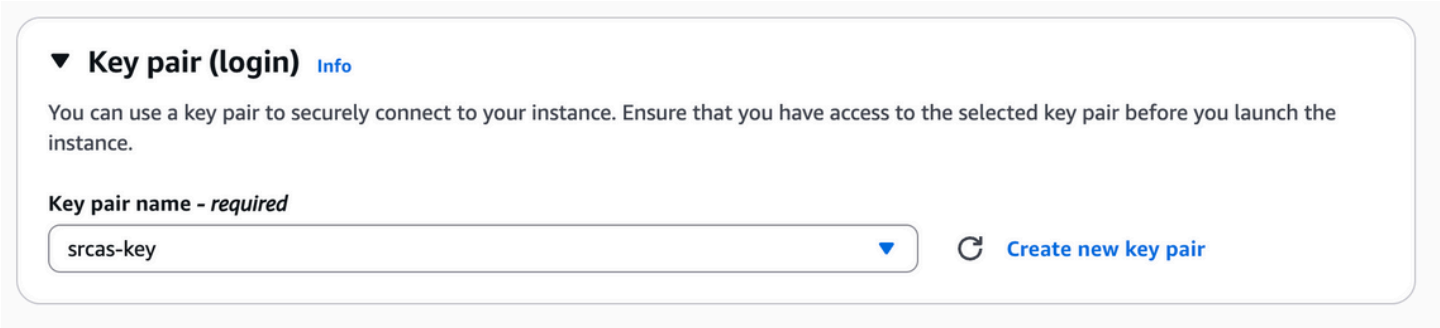
AMI : Amazon Linux kernel-6.1



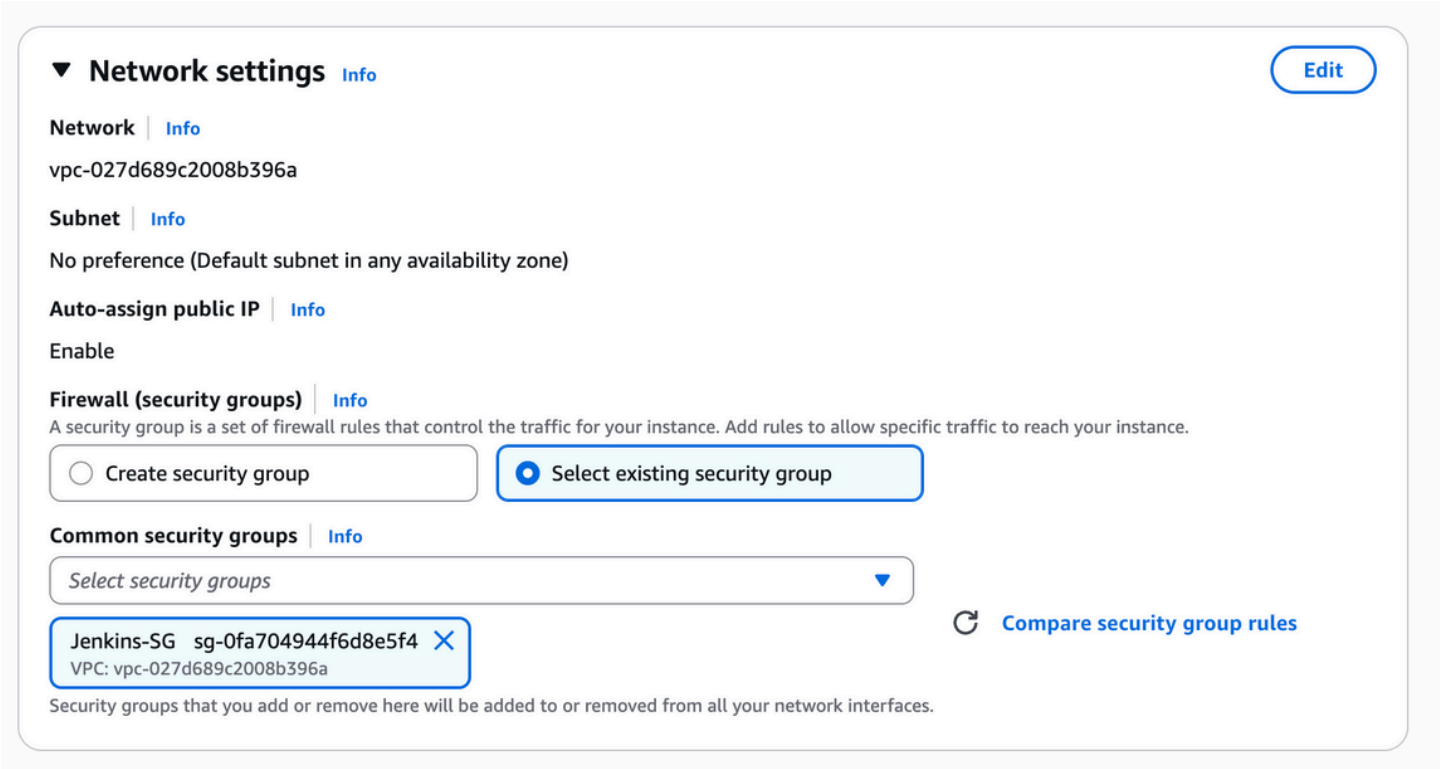
Instance Type: t3.micro



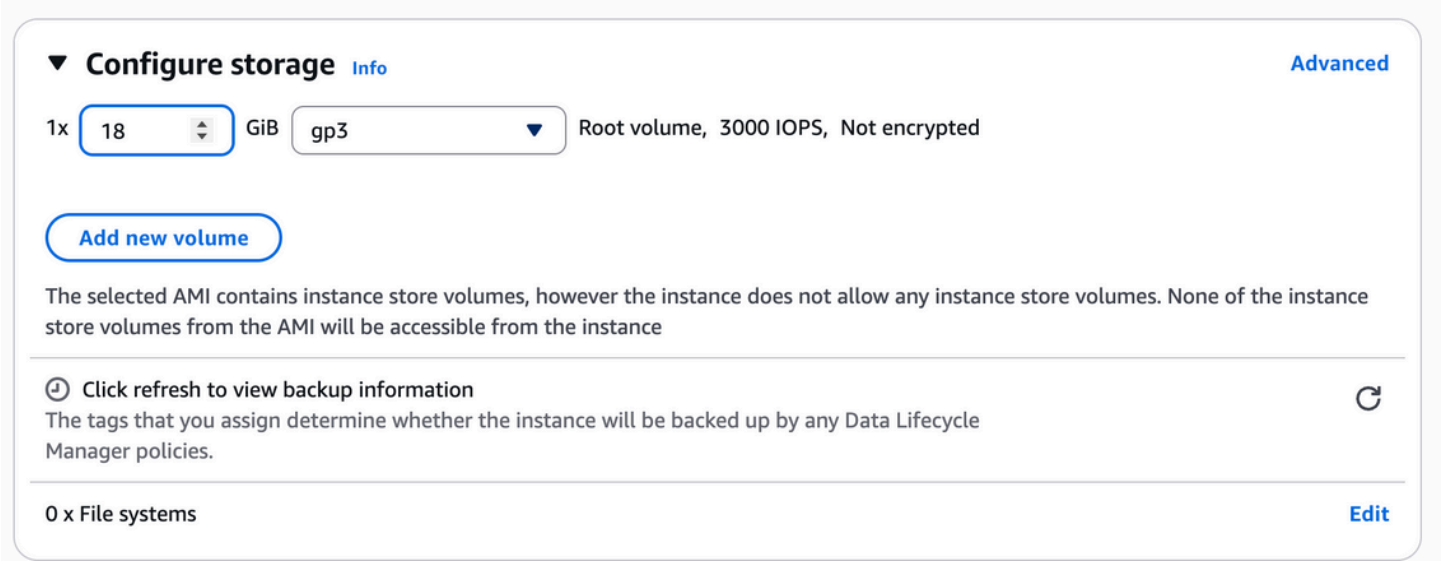
Key-pair: select existing



Security Groups : Select existing



Storage : 18 GB EBS



Now click on launch instance and then connect it.

Now login as root user : **sudo -i**

copy the below commands:

#!/bin/bash

- *sudo yum update –y*
- *sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo*
- *sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key*
- *sudo yum upgrade*
- *sudo yum install java-17-amazon-corretto -y*
- *sudo yum install jenkins -y*
- *sudo systemctl enable jenkins*
- *sudo systemctl start jenkins*
- *sudo systemctl status jenkins*
- sudo mkdir -p /var/tmp_disk
- sudo chmod 1777 /var/tmp_disk
- sudo mount --bind /var/tmp_disk /tmp
- echo '/var/tmp_disk /tmp none bind 0 0' | sudo tee -a /etc/fstab
- sudo systemctl mask tmp.mount
- df -h /tmp
- sudo systemctl restart jenkins

Reference: *all-setups/jenkins.sh at master · devops0014/all-setups*

*Execute the above commands one by one or use **.sh** file*

create a file : **vim jenkins.sh**

go to insert mode : **i**

paste the commands

now press **esc** key

and then save it (**:wq**)

to execute the file : **sh jenkins.sh**

then automatically jenkins will gets installed in your ec2 instance

After installing the Jenkins copy the public IP of your instance and then access it with 8080 port

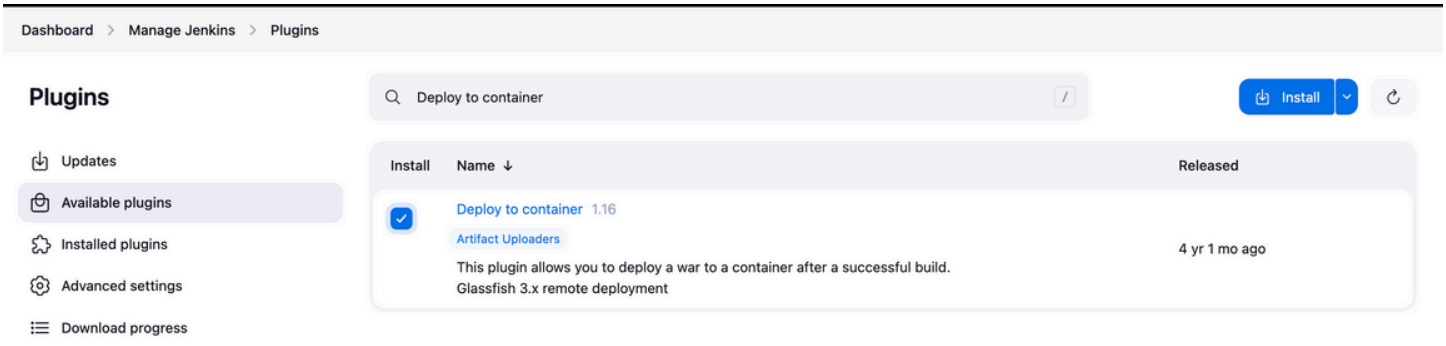
ex: 123.45.67.9:8080

Next step is setup the jenkins dashboard

Now Jenkins is Ready, Lets integrate it with GIT

Now Install **Deploy to container** plugin

Go to Manage Jenkins » Plugins » Available Plugins » Deploy to container



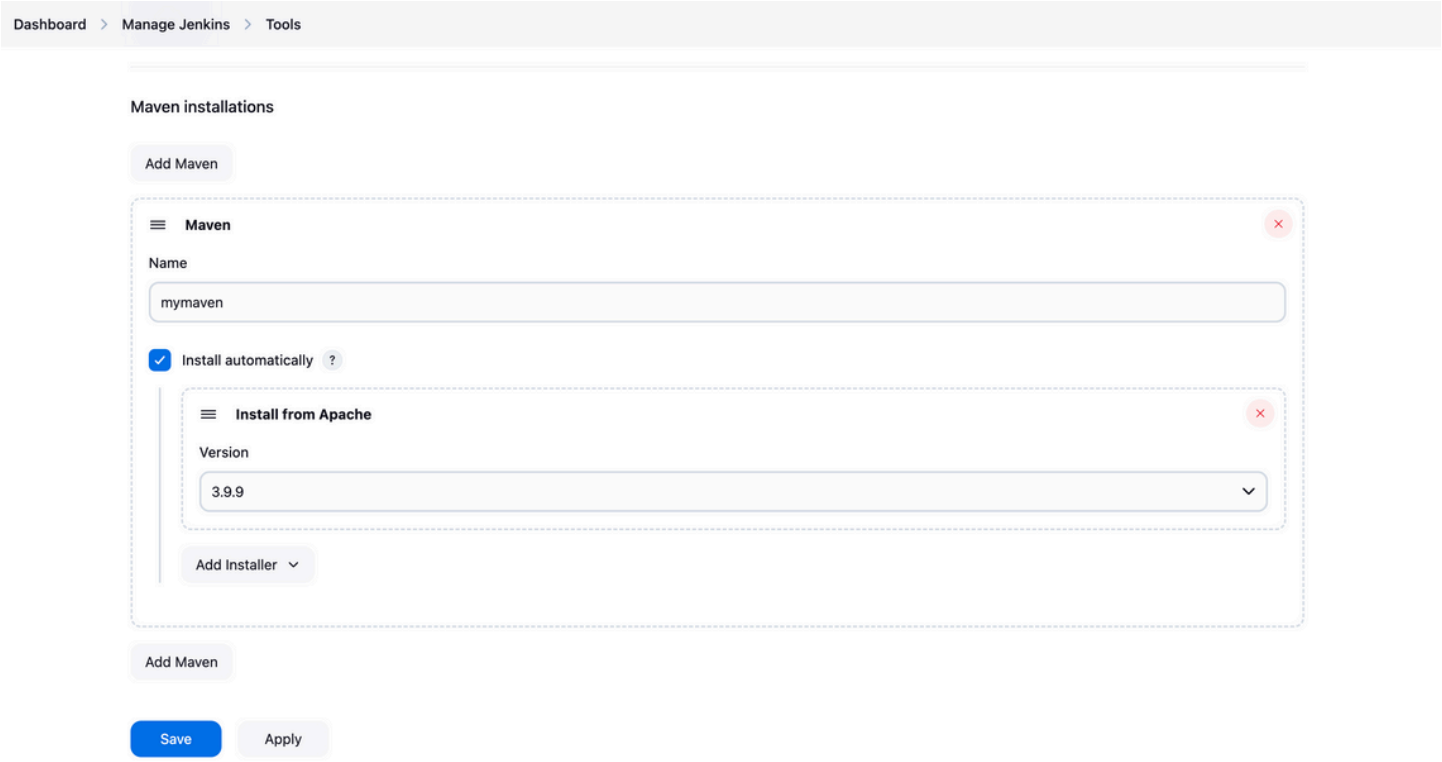
STEP-4: Integrate Maven

Go to Manage Jenkins » tools and select maven

under the **Maven Installations** click on **Add Maven**

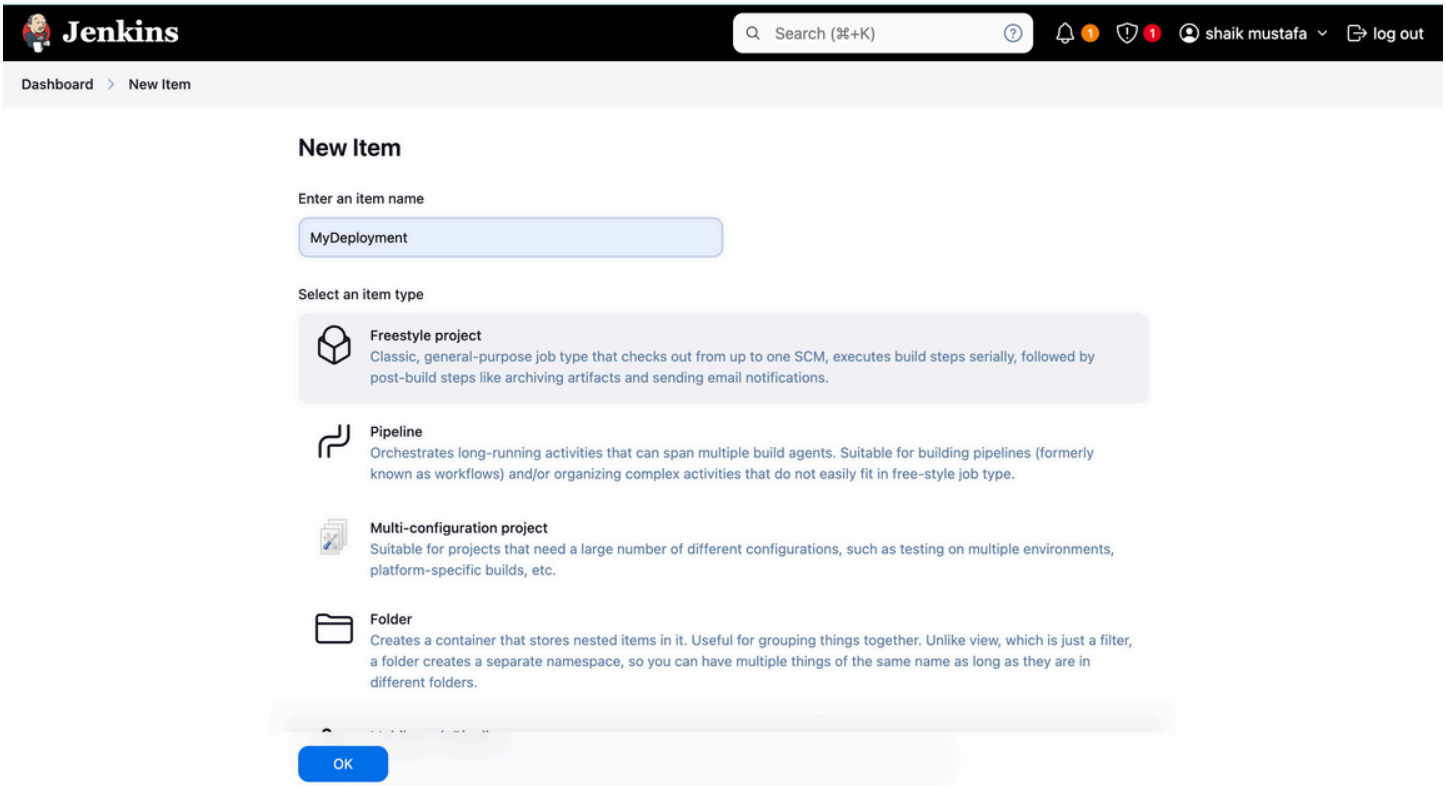
Name: **mymaven**

Version: **default (3.9.9)**

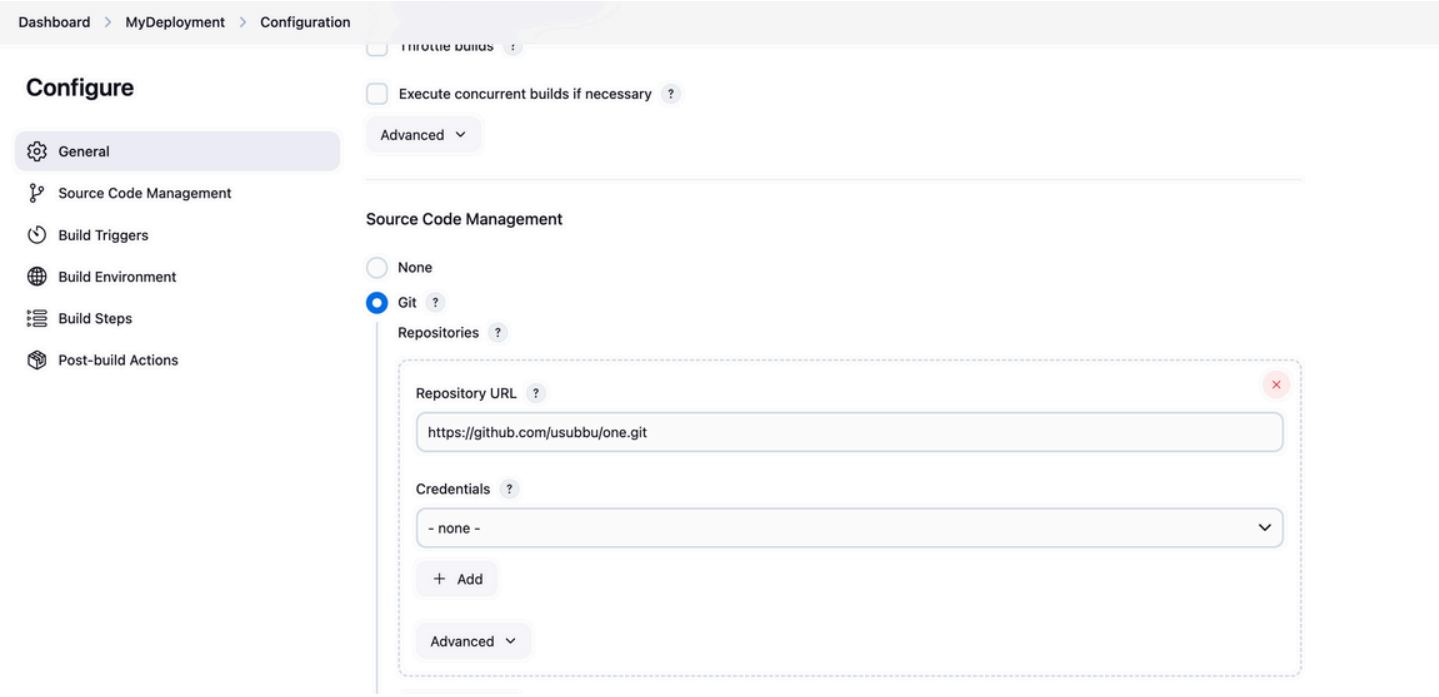


Now click on **save**

STEP-5: Create a Free Style Job

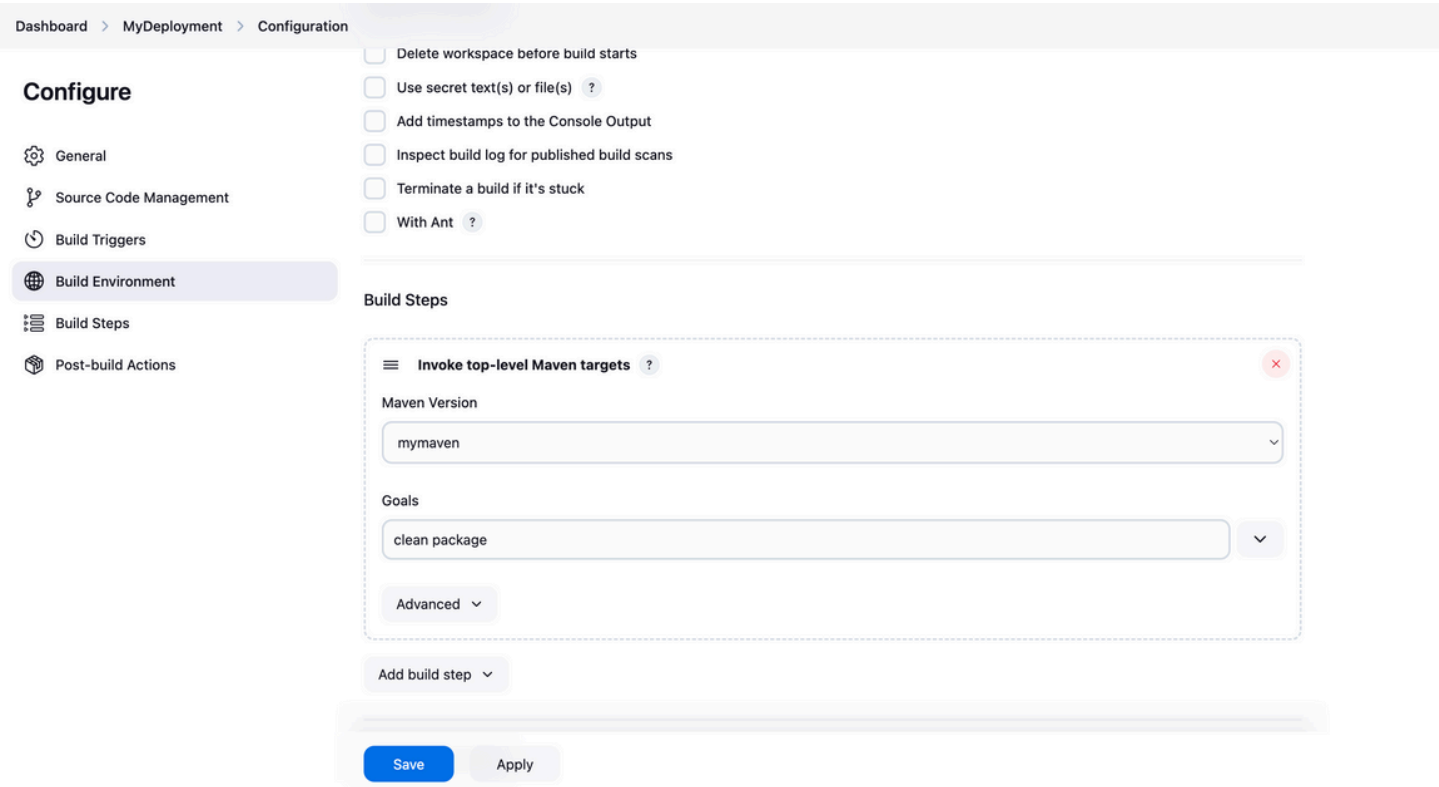


Go to **Source Code Management** and select **Git** and enter the repo-url (https://github.com/usubbu/one.git)

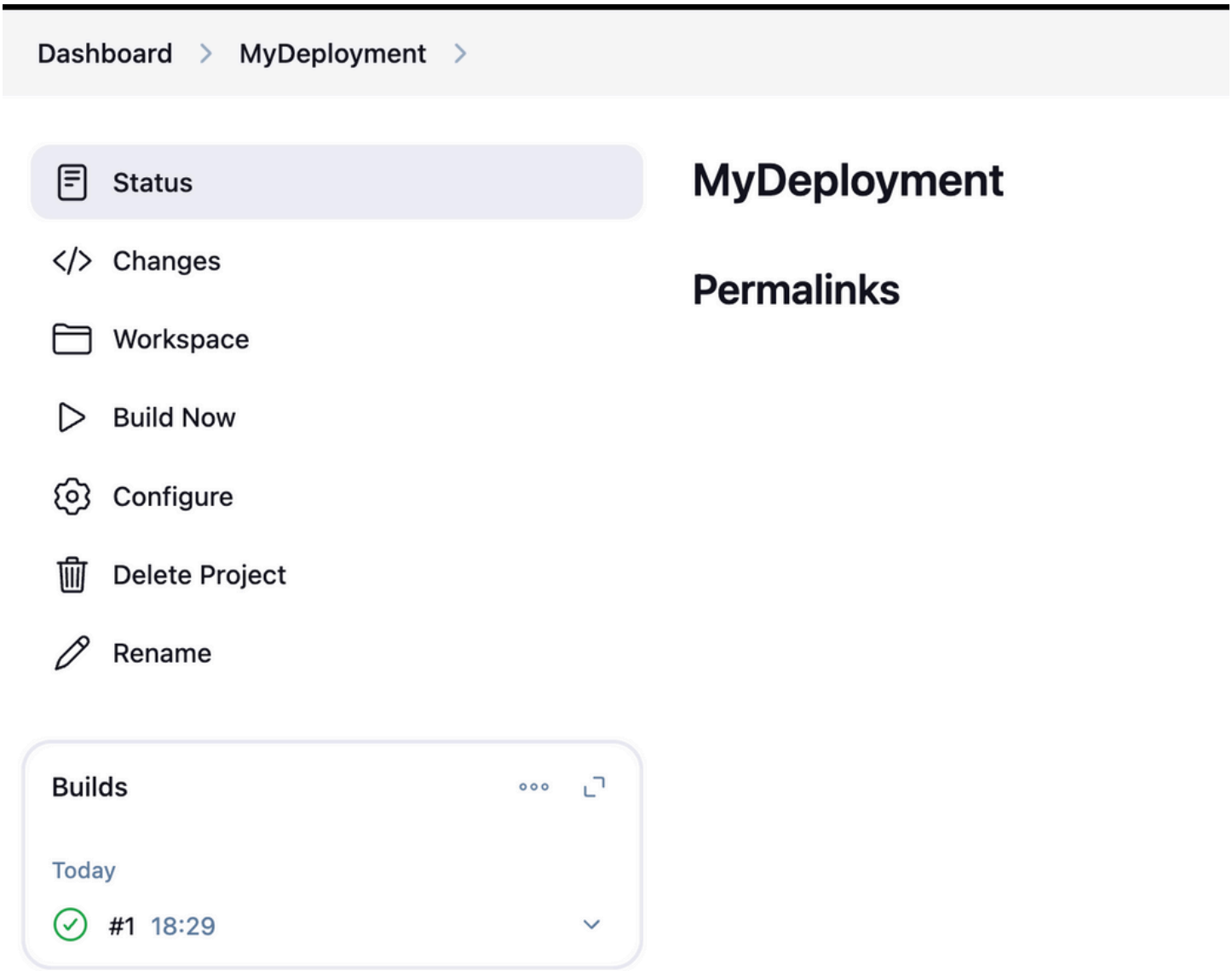


Go to **Build Steps** and select **Invoke top-level Maven targets** and select the Maven version as **mymaven**

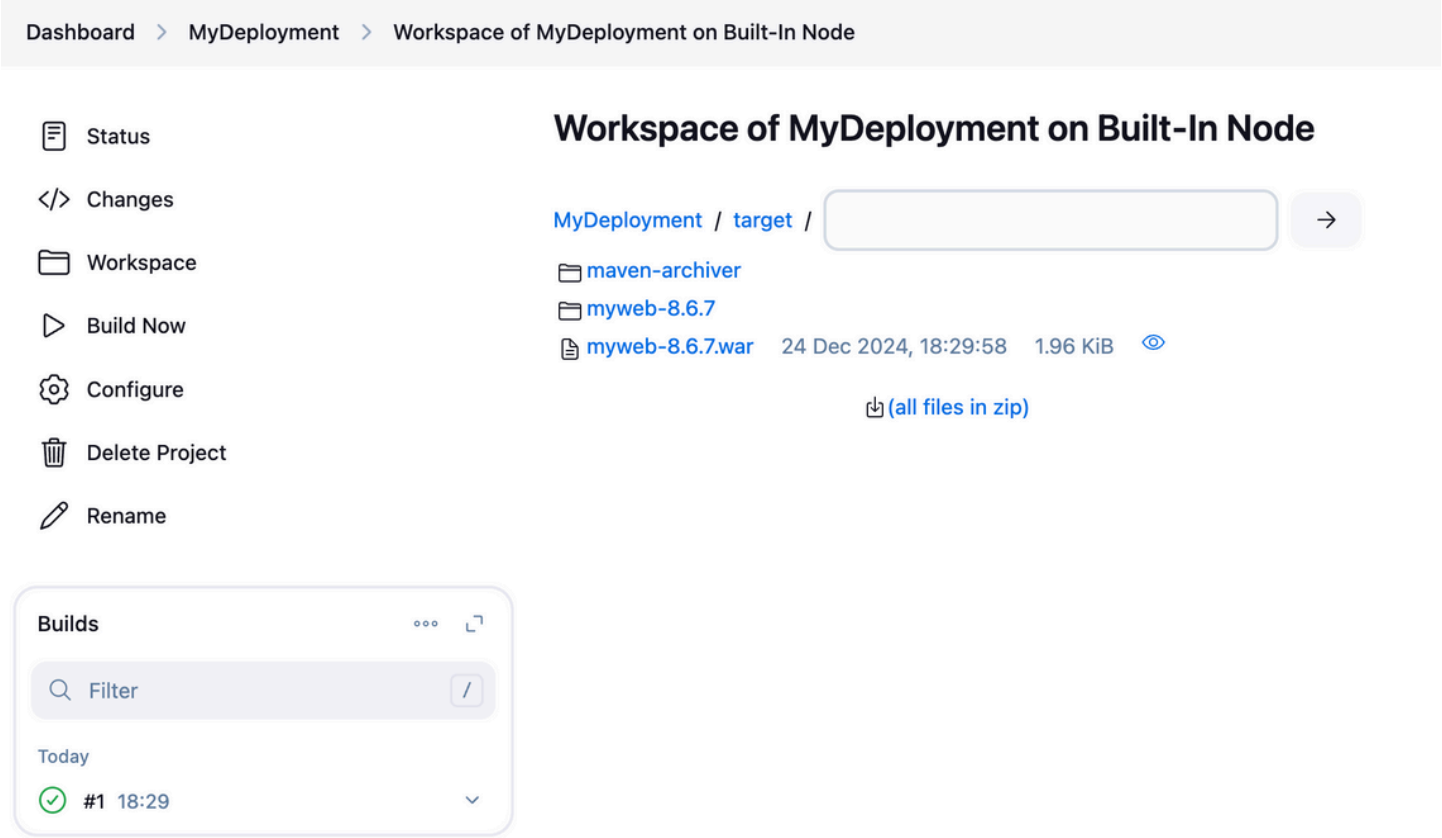
Now enter the goal as **clean package**



Save the job and click on **Build Now**

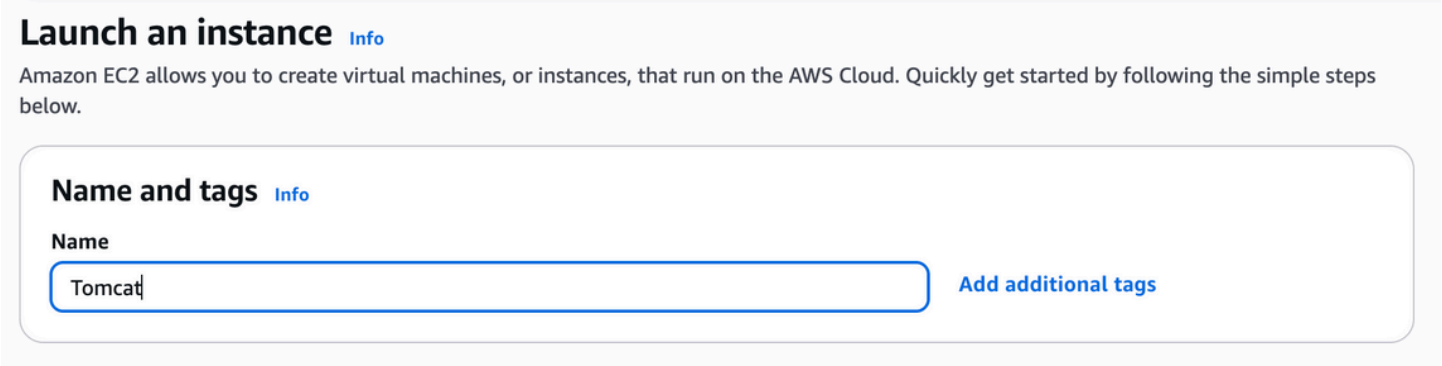


If the build gets success, go to **workspace** and open the **target** folder, we will get **war** file

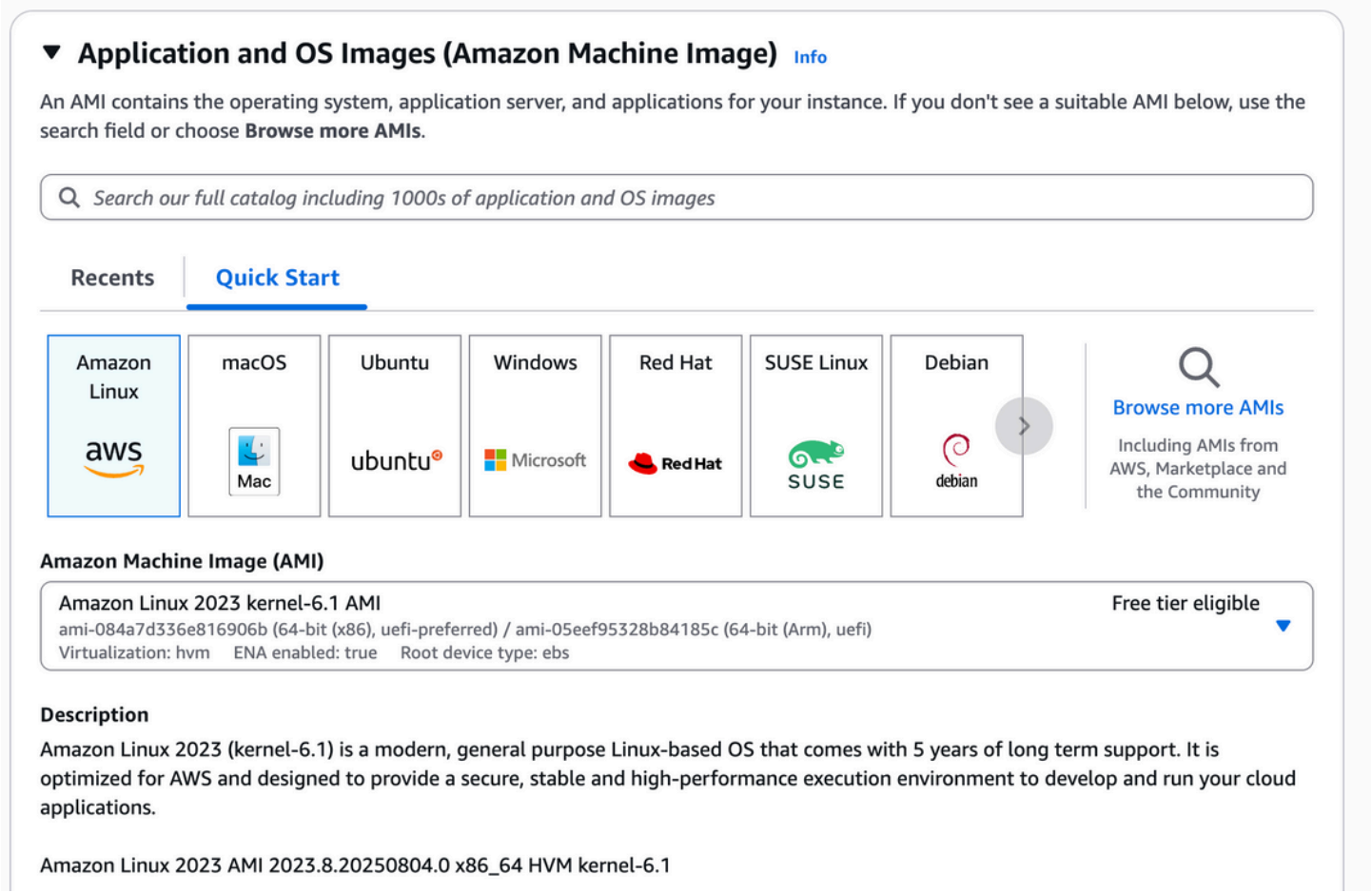


Now Launch one more instance for tomcat

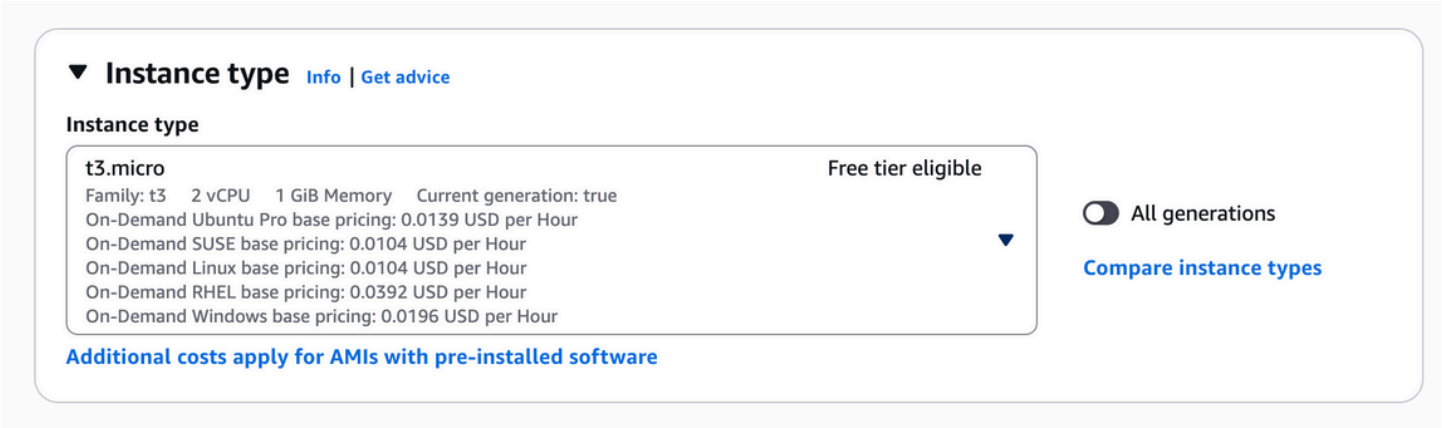
NAME : Tomcat



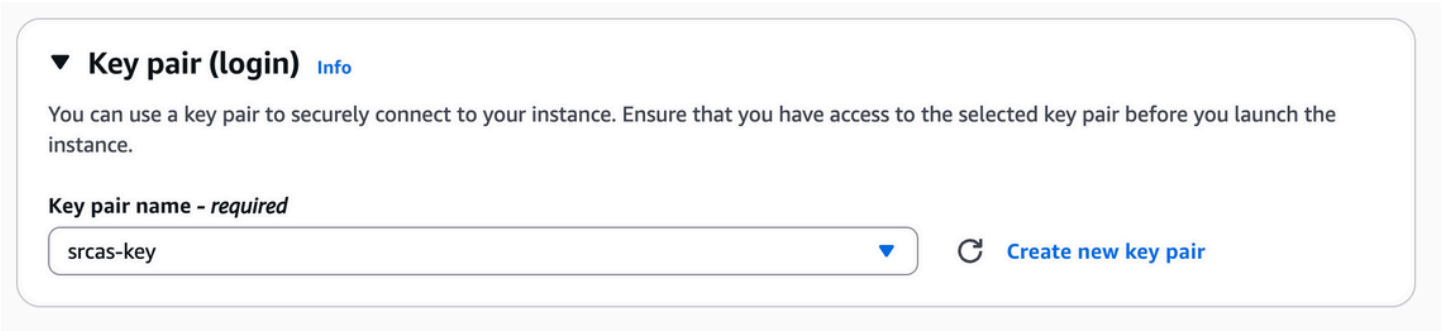
AMI : Amazon Linux Kernel-6.1



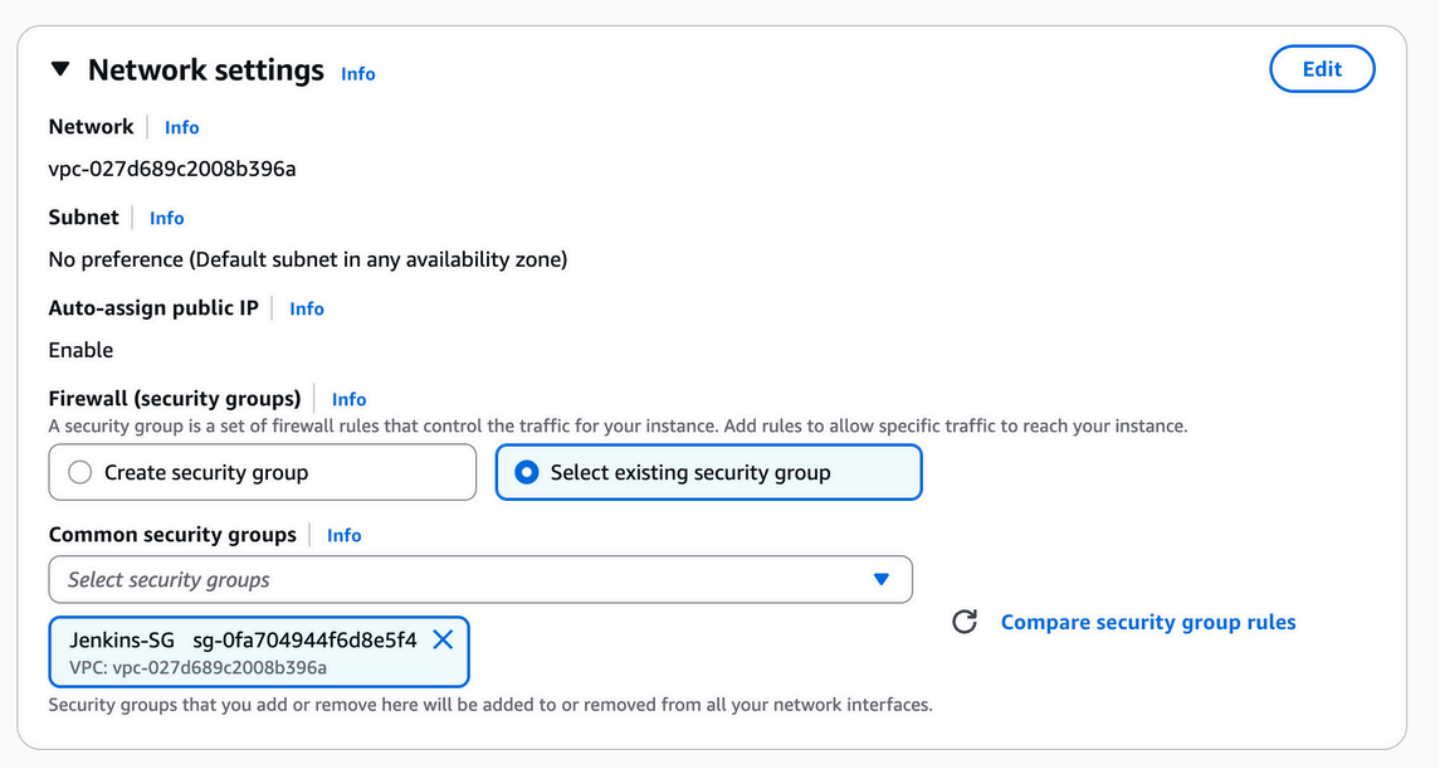
Instance Type: t3.micro



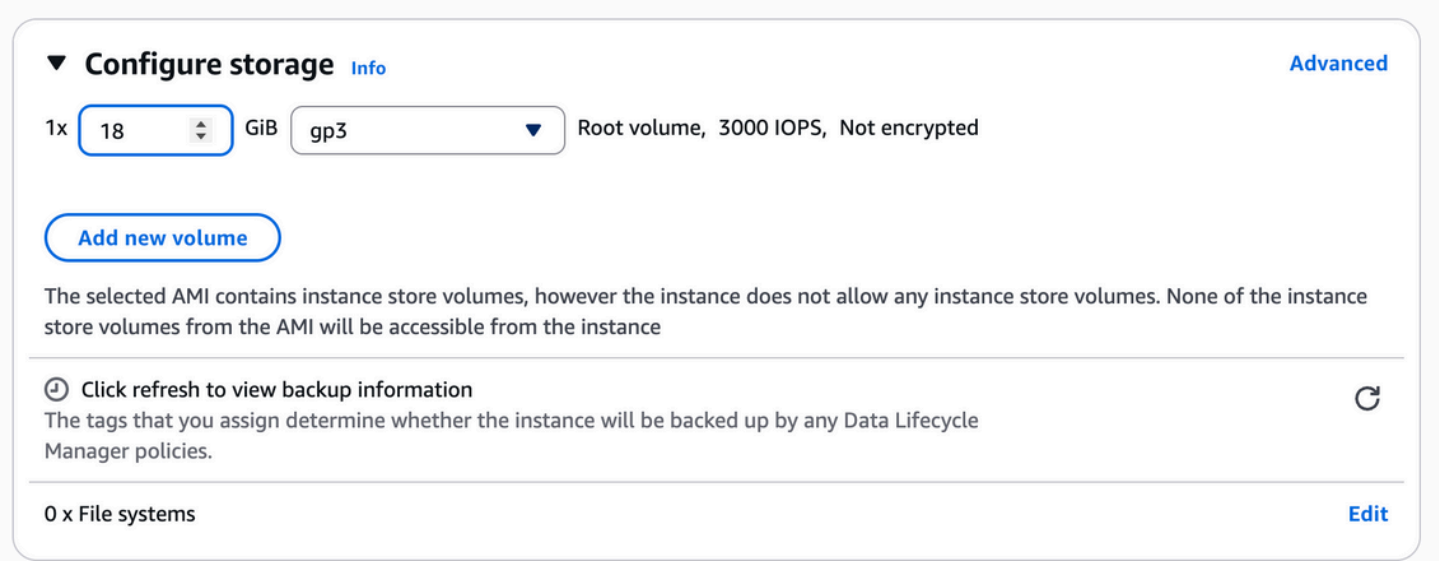
Key-pair: select existing



Security Groups : Select existing



Storage : 18 GB EBS



Now click on launch instance and then connect it.

Now login as root user : **sudo -i**

copy the below commands:

- yum install java-17-amazon-corretto -y
- wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.108/bin/apache-tomcat-9.0.108.tar.gz
- tar -zxvf apache-tomcat-9.0.108.tar.gz
- sed -i '56 a\<role rolename="manager-gui"/>' apache-tomcat-9.0.108/conf/tomcat-users.xml

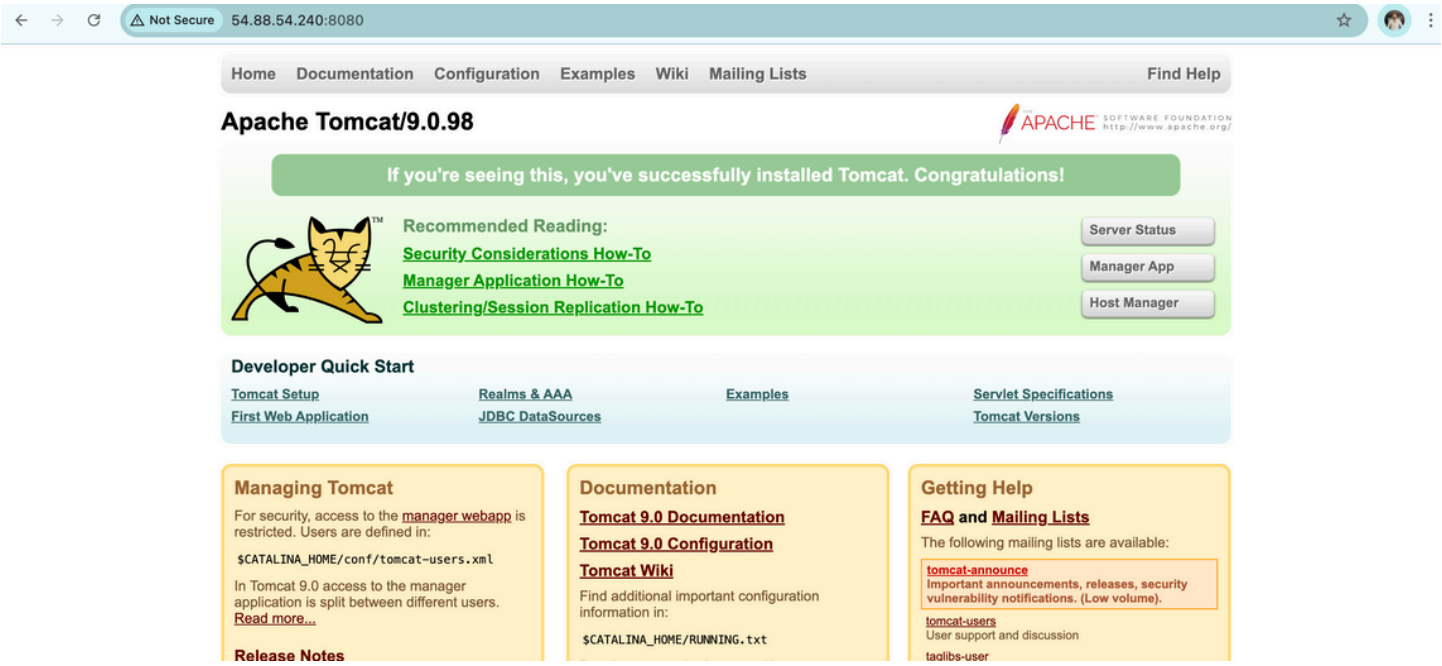
- sed -i '57 a\<role rolename="manager-script"/>' apache-tomcat-9.0.108/conf/tomcat-users.xml
- sed -i '58 a\<user username="tomcat" password="admin@123" roles="manager-gui, manager-script"/>' apache-tomcat-9.0.108/conf/tomcat-users.xml
- sed -i '59 a\</tomcat-users>' apache-tomcat-9.0.108/conf/tomcat-users.xml
- sed -i '56d' apache-tomcat-9.0.108/conf/tomcat-users.xml
- sed -i '21d' apache-tomcat-9.0.108/webapps/manager/META-INF/context.xml
- sed -i '22d' apache-tomcat-9.0.108/webapps/manager/META-INF/context.xml
- sh apache-tomcat-9.0.108/bin/startup.sh

Remember the tomcat credentials

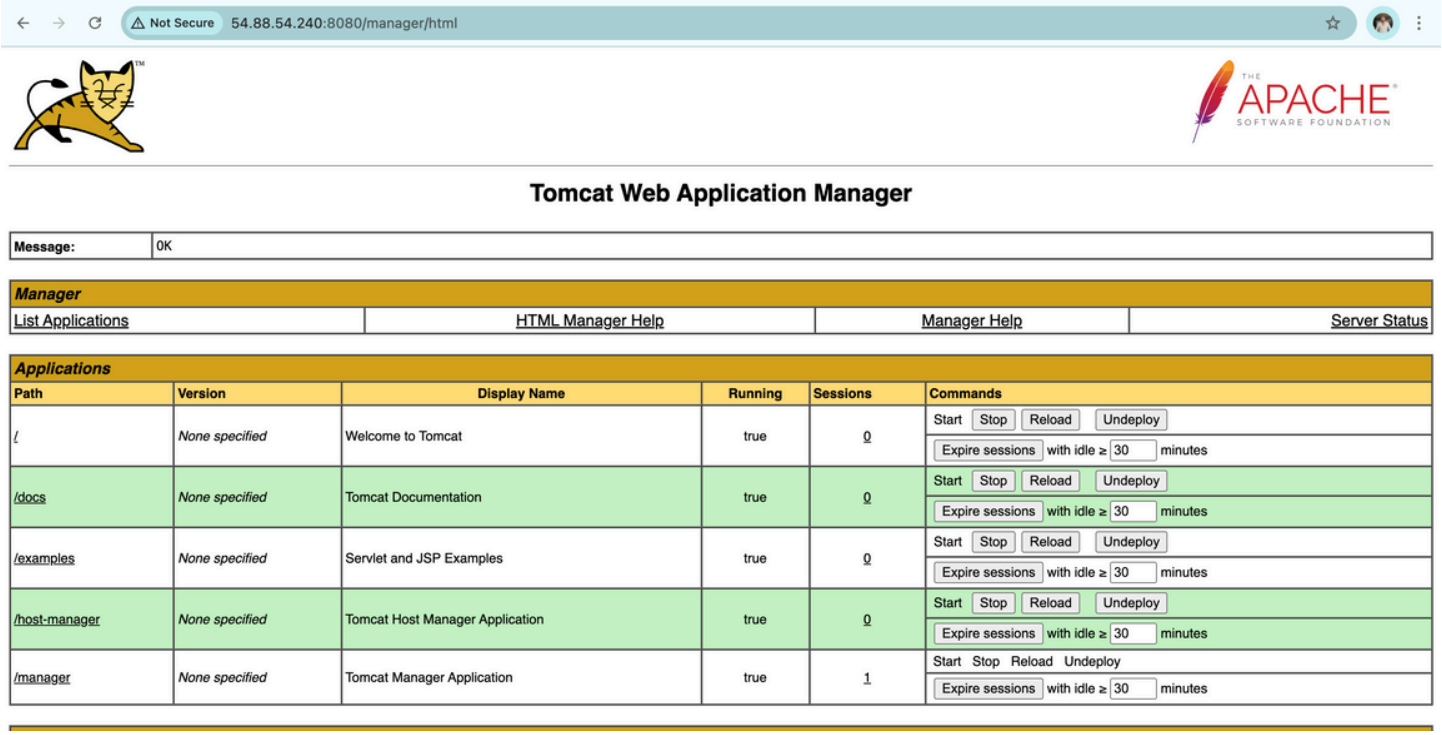
- username= **tomcat**
- password= **admin@123**

If you wish to modify, change the credentials on above script also.

Now access the tomcat dashboard with **public-ip** of tomcat server with **8080** port.



Now go to **Manager App** and enter the credentials, You will get the **Tomcat Web Application Manager** dashboard.

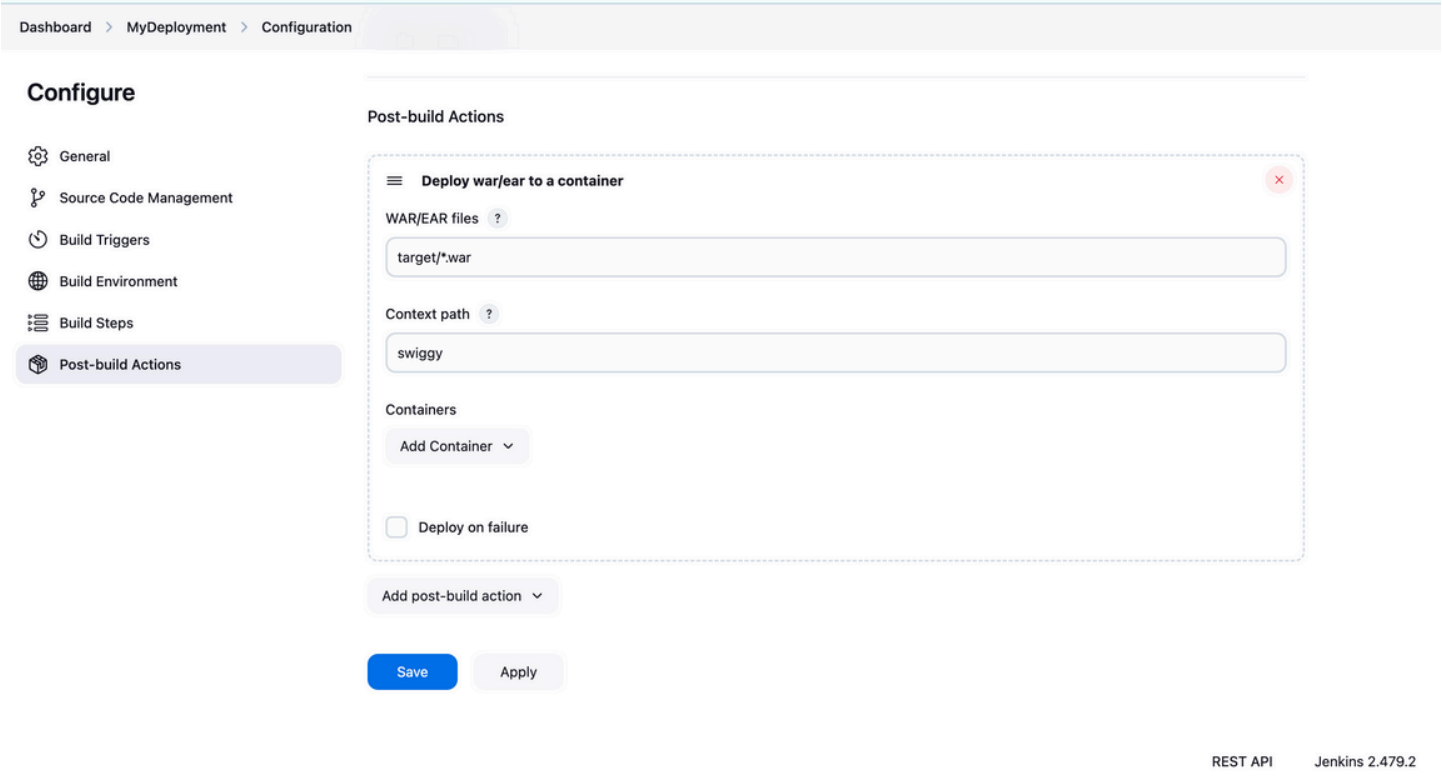


STEP-7: Integrate Tomcat with Jenkins

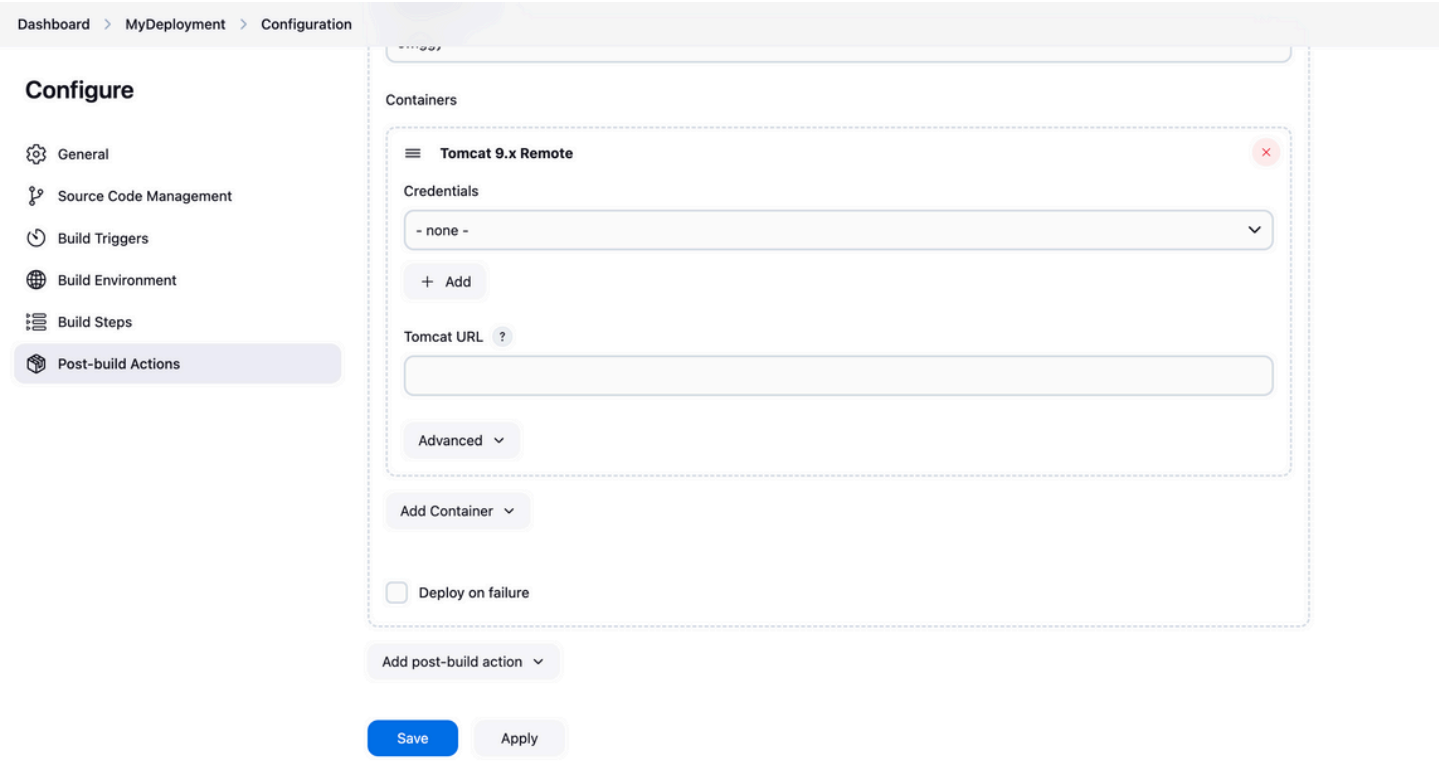
Configure the Jenkins **job** and select **Post Build Actions** and select **Deploy war/ear to a container**

WAR/EAR file: **target/*.war**

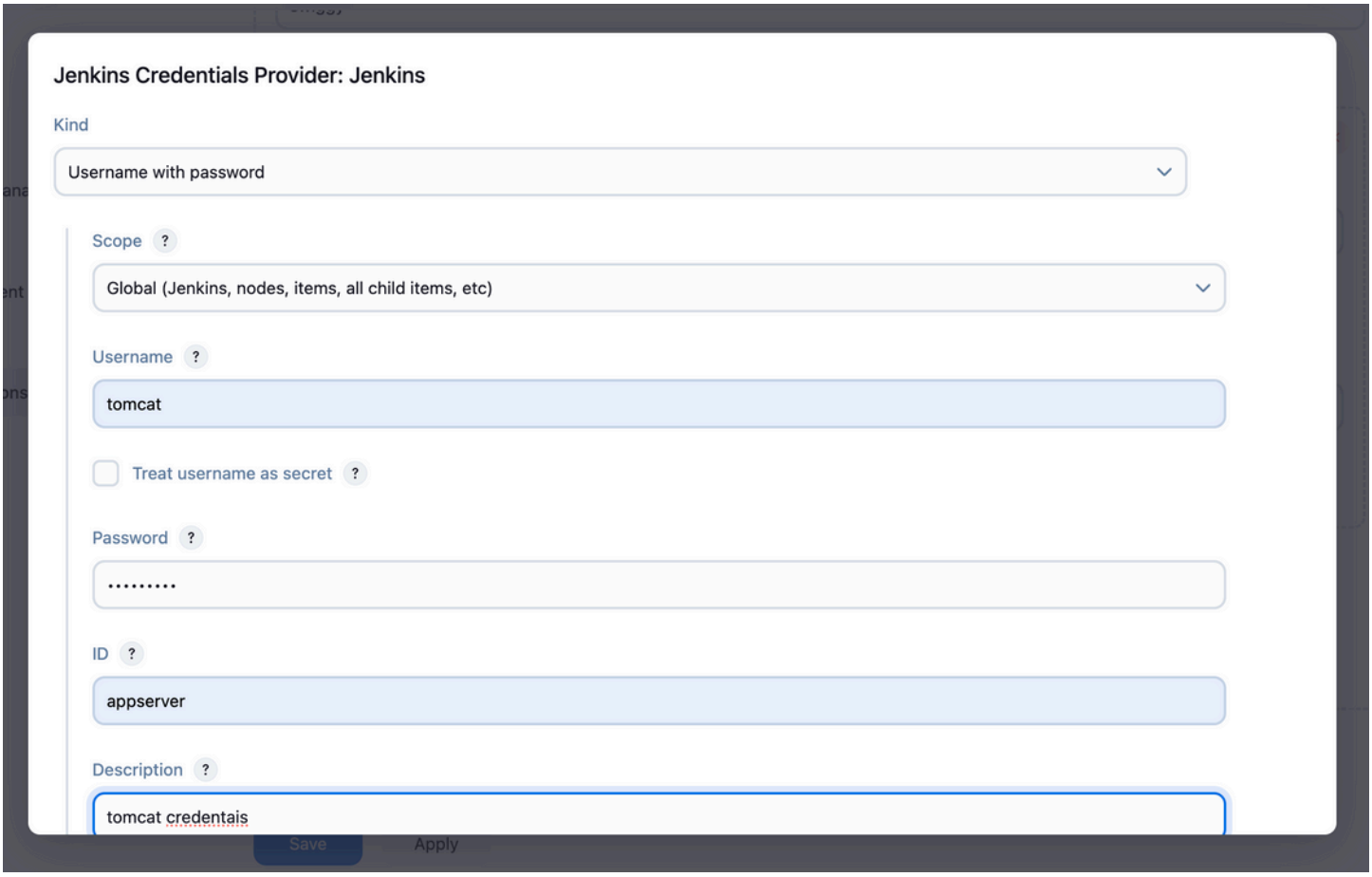
context path: **swiggy**



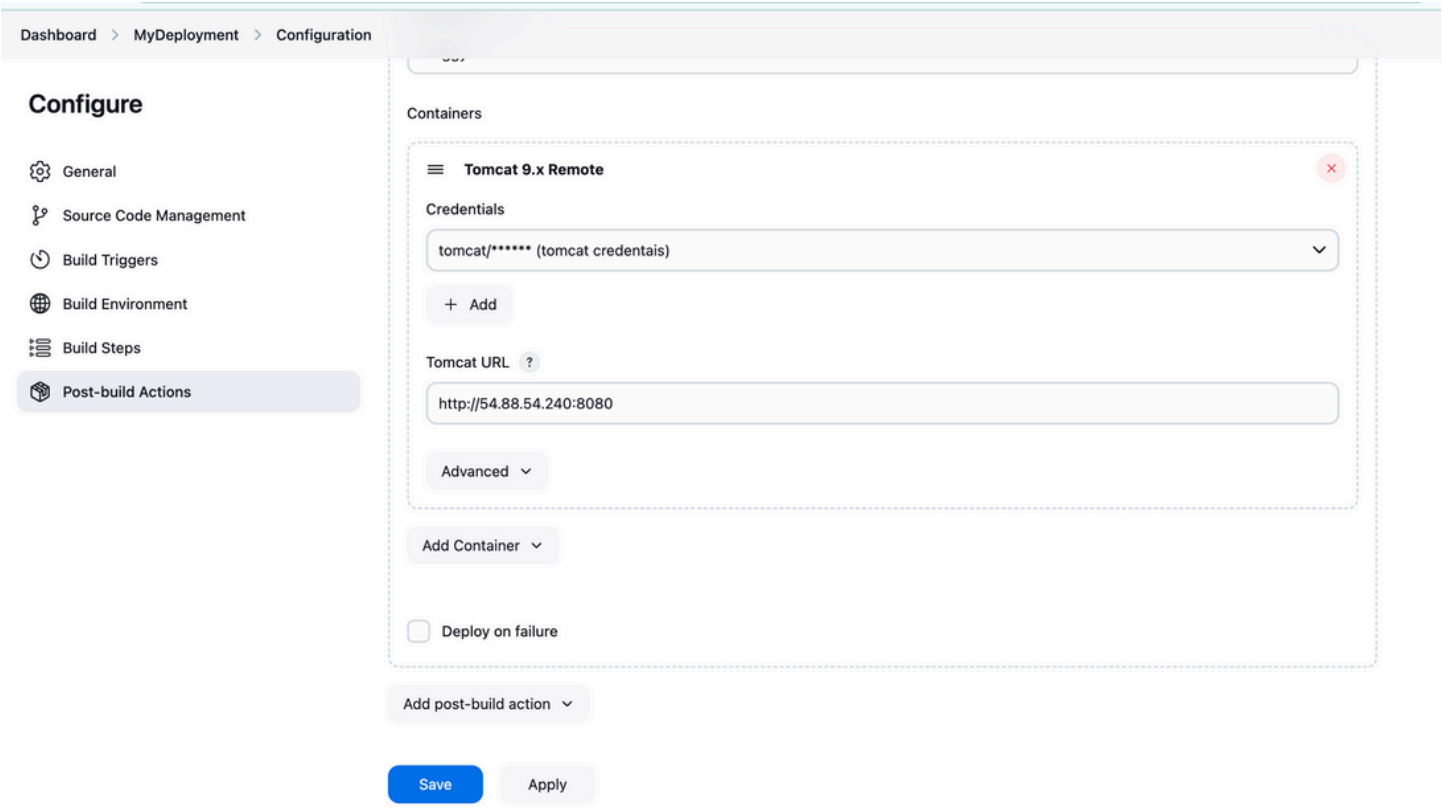
and click on **add container** and select **Tomcat 9.x Remote**



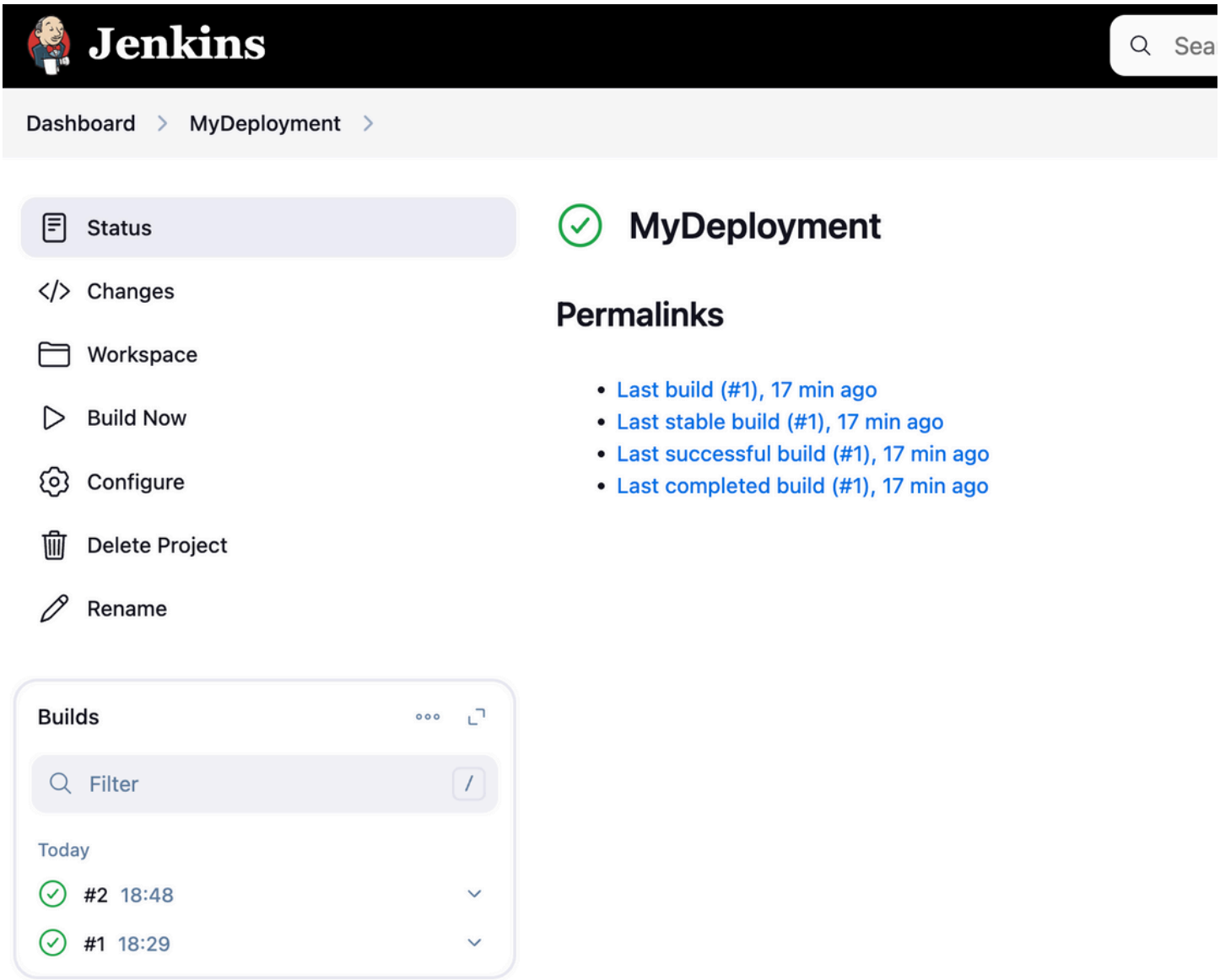
Now add the tomcat credentials, click on **Add** select **Jenkins**



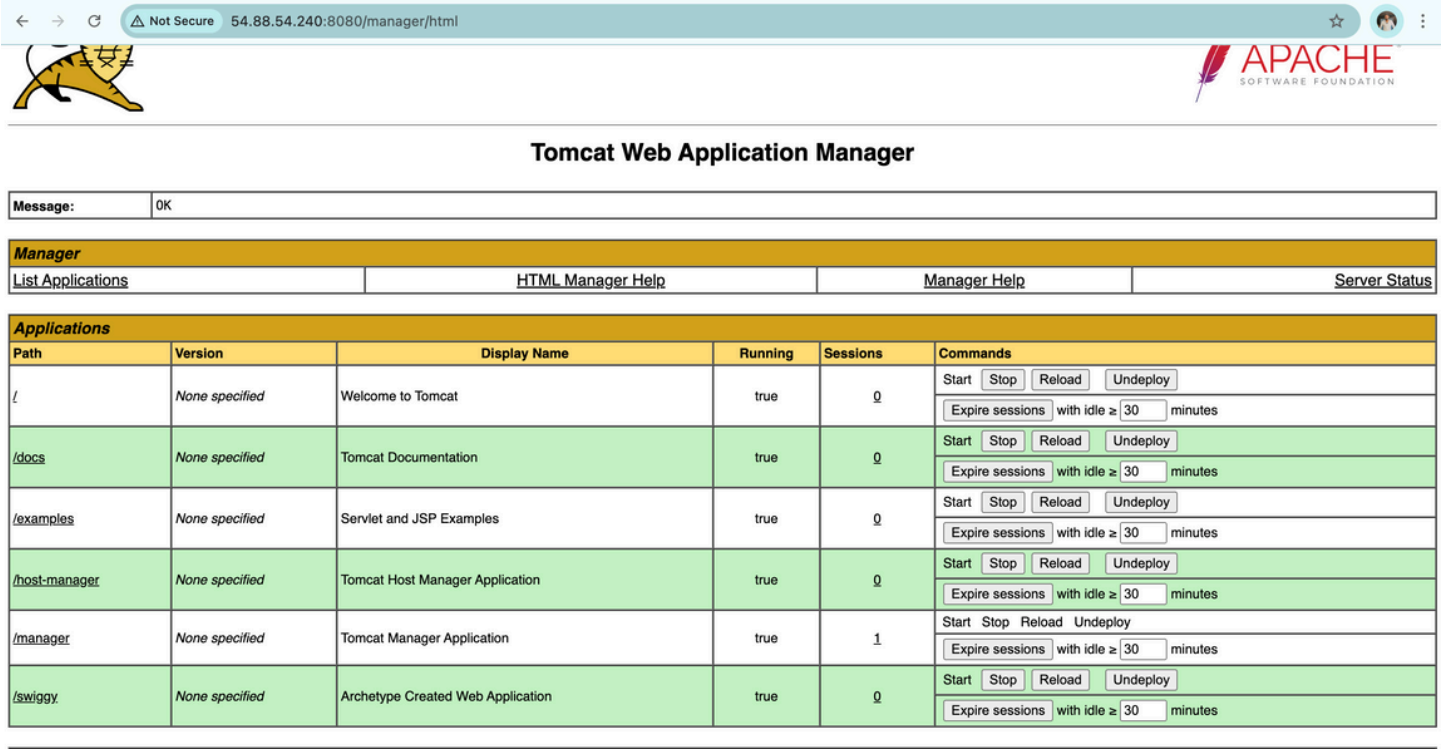
Now click on **Add** and select the **credentials** and enter the **tomcat url**



Now save the job and click on **Build Now,** If the build gets success then our application will be deployed on tomcat successfully.



The build is **success**, Now lets **refresh** the **tomcat** page and we will see **swiggy** app is **running**



Open the swiggy app and check the output.